

Week 6

Keel Bone Chicken Project

06/29/26



College of Engineering

This Week

Weekly Progress

- **Testing Updates**
 - Arranging a time to meeting with Pedro
- **Firmware Changes**
 - New Firmware flashed onto Arduino
- **Database Updates**
 - Demo of Current Progress
 - Improvements Made (1)
 - Improvements Made (2)
- **Supply Updates**
 - Initial Issue with Cabling
 - Proposed Solution
 - Purchased Solution
 - Pending Request in CIO
- **Next Week**
 - Goals for next week

Testing Updates

Arranging a time to meeting with Pedro

- As stated last week, Pedro Barbosa is a PhD candidate in Industrial Engineering (IE)
- No further authorization is needed, he just needs to let me into the facility
- Says the Piezoelectric testing of the system is feasible
- Requested a detail diagram of the system
 - Still working on this
- Likely we will meet after the 4th of July weekend (7/6)
- Will keep updates

Firmware Changes

New Firmware flashed onto Arduino

- Simple code to test all sensors connecting to database
- Reduced all unnecessary links in code to reduce bugs

```
if (central) {
  Serial.println("Connected to central");
  while (central.connected()) {
    BHY2.update(); // Update sensor fusion data

    // Check if the web app sent a new sampling rate
    if (rateChar.written()) {
      currentRate = rateChar.value();
      if (currentRate <= 0) currentRate = 1; // Prevent divide-by-zero
      Serial.print("New sampling rate: ");
      Serial.println(currentRate);
    }

    // Timer to control sampling rate
    if (millis() - lastUpdate >= (1000 / currentRate)) {
      lastUpdate = millis();

      // Format: AccelX,AccelY,AccelZ,Temp,GPI03
      String payload = String(accel_x()) + " " +
```

Database Changes

Demo of Current Progress

- Video of Logging: <https://www.youtube.com/watch?v=GDCnnC6DZ4>

Database Changes

Improvements Made (1)

- Logger time frame now only exports the needed data
- Device UUID is now characterized so it is easy to identify
- Previous Bugs: was calling database from all items
- New Code seen below:

```
// Endpoint 2: Export data based on timeframe
app.get('/api/export', async (req, res) => {
  try {
    const minutes = parseInt(req.query.minutes) || 30;
    const timeLimit = new Date(Date.now() - minutes * 60000); // Calculate past time

    // Find all records from timeLimit to now, sorted oldest to newest
    const data = await SensorData.find({ timestamp: { $gte: timeLimit } })
      .sort({ timestamp: 1 })
      .lean();
    res.json(data);
  } catch (err) {
    res.status(500).send(err.message);
  }
});
```

Database Changes

Improvements Made (1)

- Backend Database Package has full integrity
- All entries packaged and present

```
void loop() {
  BLEDevice central = BLE.central();

  if (central) {
    Serial.println("Connected to central");
    while (central.connected()) {
      BHY2.update(); // Update sensor fusion data

      // Check if the web app sent a new sampling rate
      if (rateChar.written()) {
        currentRate = rateChar.value();
        if (currentRate <= 0) currentRate = 1; // Prevent 0
        Serial.print("New sampling rate: ");
        Serial.println(currentRate);
      }

      // Timer to control sampling rate
      if (millis() - lastUpdate >= (1000 / currentRate))
        lastUpdate = millis();

      // Format: AccelX,AccelY,AccelZ,Temp,GPI03
      String payload = String(accel.x()) + "," +
        String(accel.y()) + "," +
        String(accel.z()) + "," +
        String(temp.value()) + "," +
        String(digitalRead(3));

      sensorChar.writeValue(payload);
    }
  }
  Serial.println("Disconnected from central");
}
```

Supply Updates

Initial Issue with Cabling

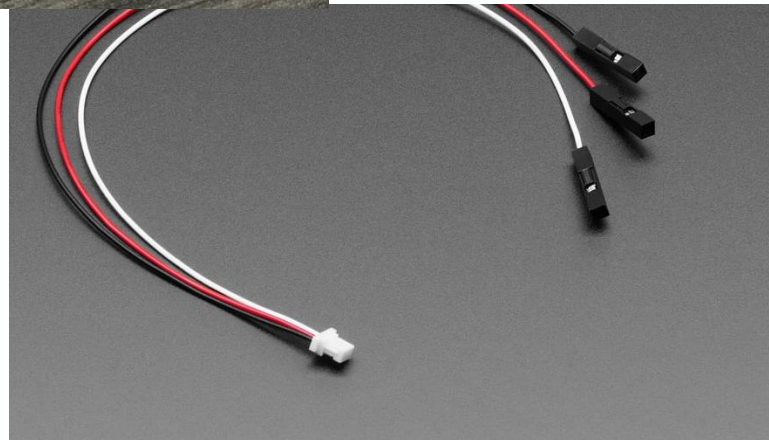
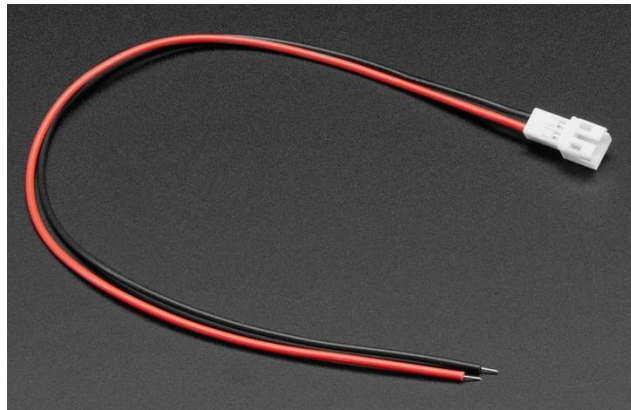
- As presented last week, the Battery is a JST PH 2 Pin **Female** Cable:
 - <https://www.digikey.com/en/products/detail/jameco-electronics/LJ503035/25966108?s=N4lgTCBcDaIDIEsAKB7ABAVgAwGZcZAF0BfIA>
- The cable needs to connect a JST PH 2 Pin **Male** → Single Pin DuPont Casing
 - These do not commercially exist



Supply Updates

Proposed Solution

- A lot of these wires exist, but none fit the given specification
- Seen below, one of these ends is always wrong or not specified with the correct header casing
- One approach is to use the right picture and connect a 3-pin male header → 2 pin male header, as seen below, but we would lose a lot of signal integrity



There are multiple versions of this item. Please select one from the options below:

Socket Headers - 100mm	\$1.50
Socket Headers - 200mm	\$1.75
Plug Headers - 100mm	Out of stock
Plug Headers - 200mm	\$1.75
Plug-Plug - 50mm	Out of stock
Plug-Plug - 100mm	Out of stock
Plug-Plug - 200mm	Out of stock

In stock

[Order now to ship today](#)

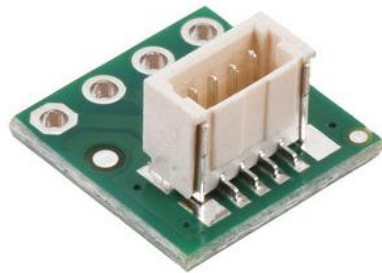
1

[Add to Cart](#)

Supply Updates

Purchased Solution

- What I think is the best solution is to use an alternative cable with no adverse effects
- The GPIO header on most boards uses the same connector, so we can take advantage of that
- A of the DuPont header to GPIO 2 JST Pin Male Header exists, which is exactly what we need
- We might run into signal integrity issues, but there will be no connectivity issues. This version is also short (as the Agriculture department requested)



Supply Updates

Pending Request in CIO

- Purchase Links
 - 2 Pin Male JST to 2 Separate Female Headers (10)
 - 2-PIN Male JST to 2 Wires (10)
 - Female Header to Female Header 5 cm (20)

ID ↕	TITLE ↕	ACCT/DEPT ↕	SERVICE ↕
1534919	<u>Supplies for Wiring</u>	WL Polytechnic (62081100)	Purchasing Center Services – Business Office
1534916	<u>Supplies for Biomedical PCB</u>	WL Polytechnic (62081100)	Purchasing Center Services – Business Office

- Requested through Purdue CIO (2) Requests): Waiting on Approval

Justification to project

I need this specific cable to connect the battery to the device, it is not available anywhere else: ht variant=32908405407840

Part/Item Number & Quantity/Unit & Price ⓘ

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<https://www.coolerguys.com/products/2pin-fan-to-gpio-female-dupont-adapter-cable?variant=32908405407>
Part ID: 840556073970
20
0.95 per unit

Ship to Name & Address ⓘ

Darrin M. Karcher
3042 Creighton Hall of Animal Sciences
270 South Russell Street
West Lafayette, IN 47907

Part/Item Number & Quantity/Unit & Price ⓘ

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See attached quote for Digikey prices, quantities, and part numbers

Ship to Name & Address ⓘ

Darrin M. Karcher
3042 Creighton Hall of Animal Sciences
270 South Russell Street

Need By Date

ASAP

Attach vendor quote, special purchase form (over \$10K), bid waiver (if necessary), VEN (if grant funds > \$2

Browse... 2026-06-29T000048.csv

Next Week

Goals for next week

- Finish System characterization and diagram for Pedro
- Ensure shipment is approved and gets sent to Professor Karcher
- Ideate changes for Nicla backpack

Thank You

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